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COMMON QUARTERLY EXAMINATION - 2026 Reg. No.

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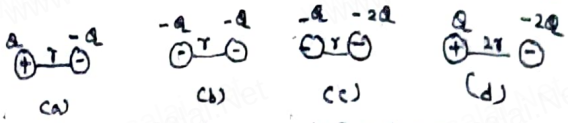
XII - PHYSICS

Time Allowed : 3.00 Hrs. Maximum Marks: 70

PART - I

I. Choose the correct answer: 15 x 1 = 15

1. Rank the electrostatic potential energies for the given system of charges in increasing order



a) 1 = 4 < 2 < 3 b) 2 = 4 < 3 < 1 c) 2 = 3 < 1 < 4 d) 3 < 1 < 2 < 2

2. Two metallic spheres of radii 1 cm and 3 cm are given charges of -1×10^{-2} C and 5×10^{-2} C respectively. If these are connected by a conducting wire, the final charge on the bigger sphere is

a) 3×10^{-2} C b) 4×10^{-2} C c) 1×10^{-2} C d) 2×10^{-2} C

3. The unit of electric potential

a) V m b) V m^{-1} c) C m d) J C^{-1}

4. In Joule's heating law, when R and t are constant, if the H is taken along the y axis and I^2 along the x axis, the graph is

a) straight line b) parabola c) circle d) ellipse

5. In a carbon resistor, the colour codes green, blue, orange and gold are given in order. The resistance of the resistor is

a) 65 KΩ, 10% b) 56 KΩ, 5% c) 65 KΩ, 5% d) 56 KΩ, 10%

6. A toaster operating at 240 V has a resistance of 120 Ω. Its power is

a) 400 W b) 2 W c) 480 W d) 240 W

7. The potential energy of magnetic dipole whose dipole moment is $\vec{P}m = \left(-0.5\hat{i} + 0.4\hat{j}\right) A\,m^2$ kept in uniform magnetic field $\vec{B} = 0.2\hat{i}\,T$

a) -0.1 J b) - 0.8 J c) 0.1 J d) 0.8 J

8. The value of gyro magnetic ratio is

a) $8.78 \times 10^{10}\ C\,kg^{-1}$ b) $9.27 \times 10^{24}\ A\,m^2$ c) $9.27 \times 10^{-24}\ A\,m^2$ d) $8.87 \times 10^{-10}\ C\,kg^{-1}$

9. When the current changes from + 2 A to - 2 A in 0.05 s, an emf of 8 V is induced in a coil. The co-efficient of self-induction of the coil is

a) 0.2 H b) 0.4 H c) 0.8 H d) 0.1 H

10. If the domestic AC supply is 230 V, 50 Hz then the peak value of voltage is

a) 230 V b) 115 V c) 325 V d) 460 V

11. Which one of the following is an electromagnetic wave?

a) α - rays b) β - rays c) Y - rays d) all of them

12. If the amplitude of the magnetic field is 3×10^{-6} T, then amplitude of the electric field for a Electromagnetic waves is

a) 100 V m^{-1} b) 300 V m^{-1} c) 600 V m^{-1} d) 900 V m^{-1}

13. Which of the following electromagnetic radiations is used for viewing objects through fog

a) micro waves b) gamma rays c) X - rays d) infra red

14. Which of the following is false for electromagnetic waves?

a) transverse b) non-mechanical waves

c) longitudinal d) produced by accelerating charges

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15. Stars twinkle due to
a) reflection b) total internal reflection c) refraction d) polarisation

PART - II

II. Answer any 6 questions. (Q.No.24 is compulsory) **6 x 2 = 12**

16. Define - electric flux.
17. Define - electrical resistivity.
18. What is meant by hysteresis?
19. State Lenz law.
20. Calculate the number of electrons in one coulomb of negative charge.
21. What are Fraunhofer lines?
22. Mention any two uses of infra red rays.
23. What are the conditions for total internal reflection?
24. Compute the magnitude of the magnetic field of a long, straight wire carrying a current of 1 A at a distance of 1 m from it. Compare it with Earth's magnetic field.

PART - III

III. Answer any 6 questions. (Q.No.32 is compulsory) **6 x 3 = 18**

25. Obtain the expression for the electric potential due to a point charge.
26. An electronics hobbyist is building a radio which requires 150 Ω in her circuit. But she has only 220 Ω, 79 Ω and 92 Ω resistors available. How can she connect the available resistors to get the desired value of resistance?
27. Discuss the conversion of galvanometer into a ammeter.
28. Derive an expression for energy stored in an inductor.
29. Find out the phase relationship between voltage and current in an A.C. circuit with a resistor.
30. Write the properties of electromagnetic waves. (any six)
31. Write the applications of Seebeck effect.
32. The self inductance of an air core solenoid is 4.8 m H. If its core is replaced by iron core, then its self-inductance becomes 1.8 H. Find out the relative permeability of iron.
33. Derive the relation between F and R of a spherical mirror.

PART - IV

IV. Answer all the questions. **5 x 5 = 25**

34. a) Calculate the electric field due to a dipole on its axial line. ***(OR)***
b) Derive an expression for phase angle between the applied voltage and current in a series RLC circuit.
35. a) Obtain the macroscopic form of Ohm's law from its microscopic form and discuss its limitation. ***(OR)***
b) Deduce the relation for the magnetic field at a point due to an infinitely long straight conductor carrying current.
36. a) Discuss the working of cyclotron in detail. ***(OR)***
b) Obtain the condition for bridge balance in Wheatstone's bridge.
37. a) Derive mirror equation. ***(OR)***
b) Explain the construction and working of transformer.
38. a) Explain the types of absorption and spectrum. ***(OR)***
b) Explain in detail the construction and working of a Van de Graaff generator.

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